Théorème du changement de variable

Si U, variable aléatoire continue

fonction g, monotone et dérivable

Si $A = g(U)$, $a = g(u)$

alors, $P_A(a) = P_U(u) \cdot \left|\frac{da}{du}\right|^{-1}$

donc si $g(U) = \tanh(U)$, $U = \mathcal{N}(\mu, \sigma^2)$ alors

$$P_{A(\cdot|s)}(a) = P_{U(\cdot|s)}(u) \cdot \left|\frac{dg}{du}\right|^{-1}$$

$$\frac{dg}{du} = \frac{d\tanh(u)}{du} = 1 - \tanh^2(u)$$

$$P_{A(\cdot|s)}(a) = P_{\mathcal{N}(\mu_s, \sigma_s^2)}(u) \cdot (1 - \tanh^2(u))^{-1}$$

$$\log(P_{A(\cdot|s)}(a)) = \log p\mathcal{N}(u|\mu_s, \sigma_s^2) - \log(1 - \tanh^2(u))$$

comme $\mathcal{N}(\mu, \sigma^2) = \mu + \sigma\epsilon$ si $\epsilon \sim \mathcal{N}(0,1)$

alors $\frac{u-\mu}{\sigma} = \epsilon$ donc

$$\log(P_{A(\cdot|s)}(a)) = \frac{-1}{2}\left(\left(\frac{u-\mu}{\sigma}\right)^2 + 2\log(\sigma) - \log(2\pi)\right) - \log(1 - \tanh^2(u_i))$$

comme $\epsilon_i = \frac{u_i - \mu_i}{\sigma_i}$

$$\log(P_{A(\cdot|s)}(a)) = -\frac{1}{2}(\epsilon^2 + 2\log(\sigma) - \log(2\pi)) - \log(1 - \tanh^2(u))$$

$$\log(P_{A(\cdot|s)}(a)) = \sum_{i=1}^{k}\left[\log(\sigma_i) - \frac{\epsilon_i^2}{2} - \frac{1}{2}\log(2\pi)\right] - \sum_{i=1}^{k}\log(1 - \tanh^2(u_i))$$ pour des action à plusieurs dim

Nous voulons $\frac{\partial \log\pi}{\partial \mu_i}$ et $\frac{\partial \log \pi}{\partial \sigma}$

$P_{A(\cdot|s)}(a) = \pi(a|s)$

$$\frac{\partial \log \pi}{\partial \mu_i} = \frac{\partial}{\partial \mu_i}\Big(\underbrace{-\log(\sigma_i) - \frac{\epsilon_i^2}{2} - \frac{1}{2}\log(2\pi)}_{\text{constantes}} - \log(1-\tanh^2(u))\Big)$$

déf :

$u = \mu_i + \sigma_i \epsilon_i$

$\phi = \tanh(u)$

$\tau = 1 - \phi^2$

$$\frac{\partial \log \pi}{\partial \mu_i} = -\frac{\partial}{\partial \mu_i}\left(\log(1-\tanh^2(u_i))\right)$$

dérivée en chaîne avec u, ϕ, τ

$$\frac{\partial \log(\tau)}{\partial \tau} = \frac{1}{\tau}$$

$$\frac{\partial \log \pi}{\partial \mu_i} = -\frac{\partial \log(\tau)}{\partial \tau} \cdot \frac{\partial \tau}{\partial \phi} \cdot \frac{\partial \phi}{\partial u_i} \cdot \frac{\partial u_i}{\partial \mu_i}$$

$$\frac{\partial \tau}{\partial \phi} = \frac{\partial (1-\phi^2)}{\partial \phi} = -2\phi$$

$$\frac{\partial \phi}{\partial u} = \frac{\partial(\tanh(u))}{\partial u} = 1-\tanh^2(u)$$

$$\frac{\partial \log \pi}{\partial \mu_i} = -\frac{1}{\tau} \cdot -2\phi \cdot (1-\tanh(u)) \cdot 1$$

$$\frac{\partial u}{\partial \mu_i} = 1$$

$$\frac{\partial \log \pi}{\partial \mu_i} = \frac{2(\tanh(u)) \cdot \cancel{(1-\tanh^2(u))}}{\cancel{(1-\tanh^2(u))}}$$

$$\frac{\partial u}{\partial \sigma_i} = \epsilon$$

$$\boxed{\frac{\partial \log \pi}{\partial \mu_i} = 2\tanh(u)}$$

$$\frac{\partial \log \pi}{\partial \sigma_i} = -\frac{\partial}{\sigma_i}\Big(\log(\sigma_i) \overbrace{+ \frac{\epsilon_i^2}{2} - \frac{1}{2}\log(2\pi)}^{\text{const}}\Big) - \frac{\partial}{\partial \sigma}\log(1-\tanh^2(u))$$

$$\frac{\partial \log \pi}{\partial \sigma_i} = -\frac{\partial \log(\sigma_i)}{\partial \sigma_i} - \frac{\partial \log(\tau)}{\partial \tau} \cdot \frac{\partial \tau}{\partial \phi} \cdot \frac{\partial \phi}{\partial u_i} \cdot \frac{\partial u_i}{\partial \mu_i}$$

$$\frac{\partial \log \pi}{\partial \sigma_i} = \frac{-1}{\sigma_i} - \left(\frac{1}{\tau} \cdot -2\phi \cdot (1-\tanh(u)) \cdot \epsilon\right)$$

$$\frac{\partial \log \pi}{\partial \sigma_i} = \frac{-1}{\sigma} - \left(\frac{1}{\cancel{\tau}} \cdot -2\phi \cdot \cancel{\tau} \cdot \epsilon\right) = \frac{-1}{\sigma} + 2\phi\epsilon = \frac{-1}{\sigma} + 2\tanh(u)\cdot\epsilon$$

$$\frac{\partial \log \pi}{\partial \sigma_i} = \frac{-1}{\sigma} + 2\tanh(u)\cdot\epsilon$$

Équations Finals.

$$L_{Q_i} = \frac{1}{2}\left(Q_{\phi_i}(s,a) - y\right)^2 \leftarrow \text{à minimiser}$$

$$y = r + \gamma(1-d)\left(\min_{j=1,2} Q_{tarr,j}(s',\tilde{a}') - \alpha \log \pi_\theta(\tilde{a}'|s')\right), \tilde{a}' \sim \pi_\theta(\cdot|s')$$

$$a_\theta(s) = \tanh(\mu_\theta(s) + \sigma_\theta(s) \odot \epsilon), \quad \epsilon \sim N(0,I) \quad \Big| \quad u = \mu_\theta(s) + \sigma_\theta(s) \odot \epsilon$$

$$\frac{\partial L}{\partial Q_\phi(s,a)} = (Q_\phi(s,a) - y)$$

$$\log \pi_\theta(a|s) = \log P_N(u|\mu_\theta(s), \sigma_\theta(s)) - \sum_{i=1}^{K} \log(1 - \tanh^2(u_i))$$

$$L_{\pi(a|s)} = \left(\min_{i=1,2} Q_{\phi_i}(s, \tilde{a}_\theta(s)) - \alpha \log \pi_\theta(\tilde{a}_\theta|s)\right) \leftarrow \text{à maximiser}$$

$$\frac{\partial L_{\pi_\theta}}{\partial \mu_i} = \left(\min_{j=1,2} \frac{\partial Q_{\phi_j}(s,\tilde{a}_\theta(s))}{\partial \tilde{a}_\theta}\right)_i \cdot (1 - \tanh^2(u_i)) - \alpha(2 \cdot \tanh(u_i))$$

$$\frac{\partial L_{\pi_\theta}}{\partial \sigma_i} = \left(\min_{j=1,2} \frac{\partial Q_{\phi_j}(s,\tilde{a}_\theta(s))}{\partial \tilde{a}_\theta}\right)_i \cdot (1 - \tanh^2(u_i)) \cdot \epsilon_i - \alpha\left(\frac{-1}{\sigma_i} + 2\tanh(u_i) \cdot \epsilon_i\right)$$

$$L_\alpha = -\alpha \log \pi(a|s) - \alpha \bar{H} \leftarrow \text{à minimiser}$$

$$\frac{\partial L_\alpha}{\partial \alpha} = \log \pi(a|s) - \bar{H}$$

$$\theta_{tarr_i} = \tau \theta_i + (1-\tau)\theta_{tarr_i} \quad (\text{pour } i = 1,2)$$